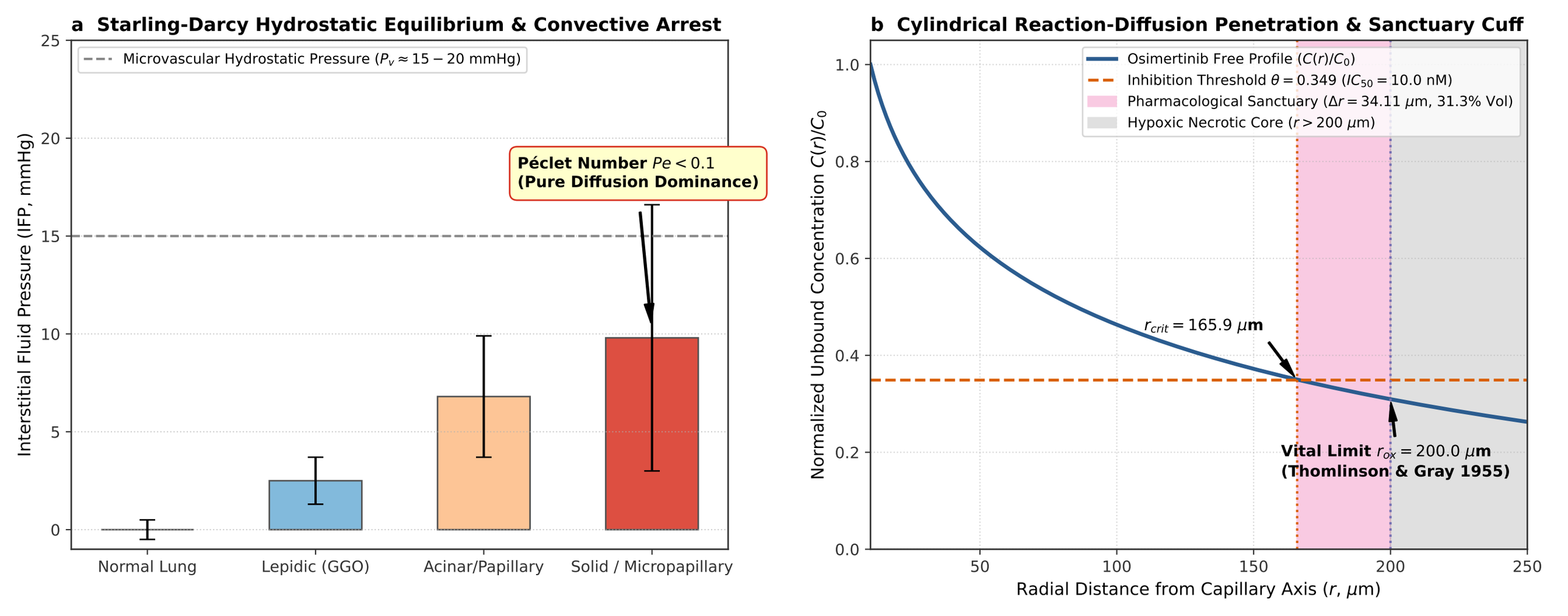
a Starling-Darcy Hydrostatic Equilibrium & Convective Arrest
Microvascular Hydrostatic Pressure ($P_v \approx 15 - 20$ mmHg)
Péclet Number $Pe < 0.1$
(Pure Diffusion Dominance)
Interstitial Fluid Pressure (IFP, mmHg)
25
20
15
10
5
0
Normal Lung
Lepidic (GGO)
Acinar/Papillary
Solid / Micropapillary
b Cylindrical Reaction-Diffusion Penetration & Sanctuary Cuff
Osimertinib Free Profile ($C(r)/C_0$)
Inhibition Threshold $\theta = 0.349$ ($IC_{50} = 10.0$ nM)
Pharmacological Sanctuary ($\Delta r = 34.11$ μm, 31.3% Vol)
Hypoxic Necrotic Core ($r > 200$ μm)
Normalized Unbound Concentration $C(r)/C_0$
1.0
0.8
0.6
0.4
0.2
0.0
$r_{crit} = 165.9$ μm
Vital Limit $r_{ox} = 200.0$ μm
(Thomlinson & Gray 1955)
50
100
150
200
250
Radial Distance from Capillary Axis (r, μm)